Unit: Unit 0: Foundations of Algebra
Topic: Rational Numbers Decimals

Name: ______________
Date: __________

Homework (Jalapeno)

Q1. The ancient Egyptians used $\frac{1}{2}$ to represent half a day. Which decimal is equivalent to $\frac{1}{2}$?
(A) 0.1
(B) 0.5
(C) 0.8
(D) 1.0
(E) 1.2

Q2. In ancient Babylon, the ratio of a circle's circumference to its diameter was approximated as 3.125. What is this value as a fraction?
(A) $\frac{25}{8}$
(B) $\frac{5}{2}$
(C) $\frac{10}{3}$
(D) $\frac{8}{5}$
(E) $\frac{3}{8}$

Q3. The Mayans used a positional notation system. If $x = 0.45$, what is x as a percent?
(A) 4
(B) 45
(C) 0.45
(D) 45.5
(E) 4.5

Q4. The ancient Greeks approximated π as 3.14. What is 3.14 as a fraction?
(A) $\frac{14}{5}$
(B) $\frac{22}{7}$
(C) $\frac{3}{10}$
(D) $\frac{5}{16}$
(E) $\frac{7}{22}$

Q5. A scroll from ancient China measured 2.5 meters. What is 2.5 meters as a fraction?
(A) $\frac{1}{2}$
(B) $\frac{2}{5}$
(C) $\frac{5}{2}$
(D) $\frac{10}{5}$
(E) $\frac{25}{10}$

Q6. Which of the following rational numbers is greatest: 0.8, $\frac{4}{5}$, 0.75, $\frac{3}{4}$, or 0.9?
(A) 0.8
(B) $\frac{4}{5}$
(C) 0.75
(D) $\frac{3}{4}$
(E) 0.9

Q7. A ancient artifact weighed 1.25 kilograms. What is 1.25 kilograms as a fraction?
(A) $\frac{1}{4}$
(B) $\frac{5}{4}$
(C) $\frac{4}{5}$
(D) $\frac{2}{5}$
(E) $\frac{5}{2}$

Q8. Which of the following decimals is equivalent to $\frac{3}{8}$?
(A) 0.35
(B) 0.37
(C) 0.375
(D) 0.38
(E) 0.4

Open Ended Questions (FRQ)

Q9. Draw a number line and plot the rational numbers -2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, and 2. Label each point.

Q10. A historian discovered an ancient text with a recipe that included $\frac{3}{4}$ cup of flour. Convert $\frac{3}{4}$ to a decimal and then to a percent. Show your work.

Chef's Tips

- Tip 1: Review the historical context of rational numbers and decimals.
- Tip 2: Emphasize the importance of converting between fractions, decimals, and percents.
- Tip 3: Encourage students to use number lines to visualize rational numbers.